* CS Stage with Diode Connected Load:

In many CMOS technologies, it is difficult to fabricate resistors with tightly-controlled values or a reasonable physical size. Consequently, it is desirable to replace R_D with a MOS transistor.

A MOSFET can operate as a small-signal resistor if its gate and drain are shorted. Called a "diode-connected" device in analogy with its bipolar counterpart, this configuration exhibits a small-signal behavior similar to a two-terminal resistor.

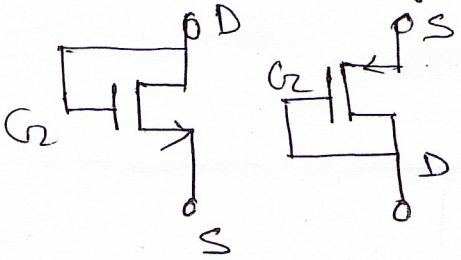
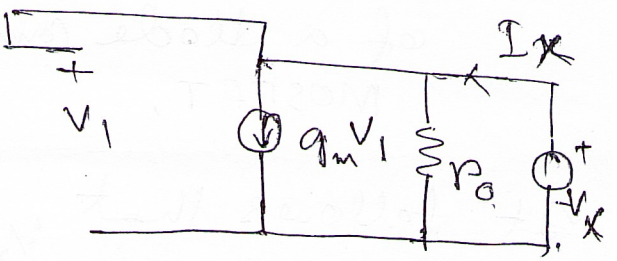


Fig. 1(a) Diode connected NMOS and PMOS devices.



(b) Small-signal equivalent circuit.

Note that the transistor is always in saturation because the drain and the gate have the same potential.

Using the small-signal equivalent Fig(b), to obtain the impedance of the ~~diode~~ device, we write $V_1 = V_X$ and $I_X = V_X/r_o + g_m V_X$. That is, the impedance of the diode is simply equal to $(1/g_m) \| r_o \approx 1/g_m$. If body effect exists, we can use the circuit in Fig. 2 to write

$$V_1 = -V_X, \quad V_{bs} = -V_X$$

and $$(g_m + g_{mb})V_x + \frac{V_x}{r_o} = I_x \quad \text{---} (1)$$

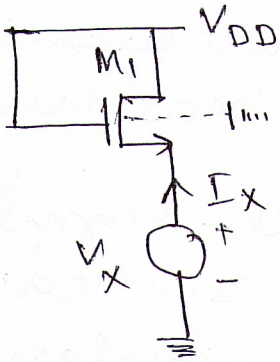
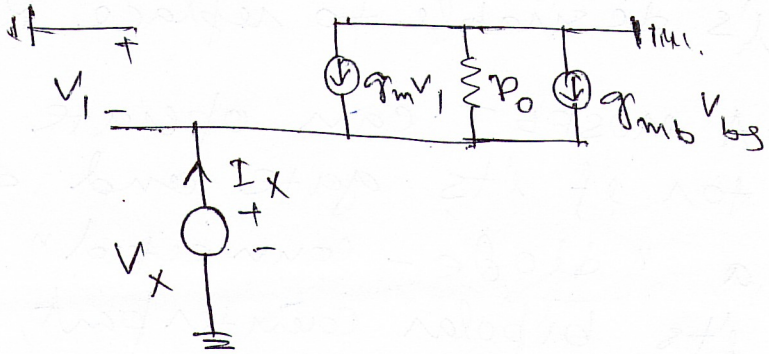


Fig. 2 (a) Arrangement for measuring the equivalent resistance of a diode connected MOSFET,



2 (b) Small-signal equivalent circuit.

It follows that

$$\frac{V_x}{I_x} = \frac{1}{g_m + g_{mb} + r_o^{-1}} \quad \text{---} (2)$$

$$= \frac{1}{g_m + g_{mb}} \parallel r_o \quad \text{---} (3)$$

$$\approx \frac{1}{g_m + g_{mb}} \quad \text{---} (4)$$

Interestingly, the impedance seen at the source of M_1 is lower when body effect is included.

We now study a common-source stage with a diode-connected load. For negligible channel length modulation (Eq. 4) can be substituted in

$A_V = -g_m R_D$ for the load impedance

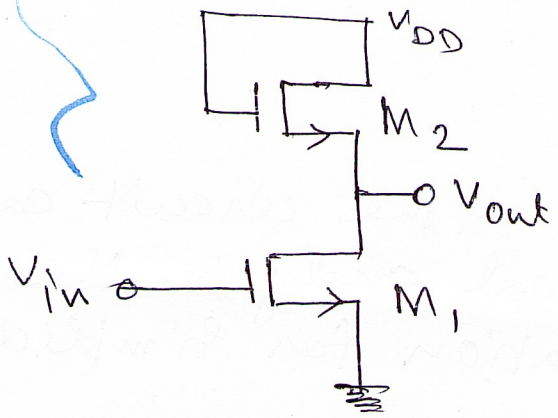


Fig: 3. CS Stage with diode connected load.

Yielding

$$A_V = -g_{m1} \cdot \frac{1}{g_{m2} + g_{mb2}} \quad \text{---} (5)$$

$$= -\frac{g_{m1}}{g_{m2}} \cdot \frac{1}{1+\eta} \quad \text{---} (6)$$

where $\eta = \frac{g_{mb2}}{g_{m2}}$. Expressing g_{m1} and g_{m2} in terms of device dimensions and bias currents we have

$$A_V = -\frac{\sqrt{2\mu_n C_{ox} (W/L)_1 I_{D1}}}{\sqrt{2\mu_n C_{ox} (W/L)_2 I_{D2}}} \cdot \frac{1}{1+\eta} \quad \text{---} (7)$$

and, since $I_{D1} = I_{D2}$

$$A_V = -\sqrt{\frac{(W/L)_1}{(W/L)_2}} \cdot \frac{1}{1+\eta} \quad \text{---} (8)$$

This equn reveals an interesting property: if the variation of η with the output voltage is neglected, the gain is independent of the bias currents and voltages.

(so long as M_1 stays in saturation). In other words, as the input and output signals levels vary, the gain remains relatively constant, indicating that the input-output characteristic is relatively linear.

The linear behavior of the circuit can also be confirmed by large-signal analysis. Neglecting channel-length modulation for simplicity, we have in Fig. 3

$$\frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 (V_{in}-V_{TH1})^2 = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_2 (V_{DD}-V_{out}-V_{TH2})^2 \quad \text{---(9)}$$

and hence

$$\sqrt{\left(\frac{W}{L}\right)_1}(V_{in}-V_{TH1}) = \sqrt{\left(\frac{W}{L}\right)_2}(V_{DD}-V_{out}-V_{TH2}) \quad \text{---(10)}$$

Thus, if the variation of V_{TH2} with V_{out} is small, the circuit exhibits a linear input-output characteristic. The small-signal gain can also be computed by differentiating both sides with respect to V_{in}

$$\sqrt{\left(\frac{W}{L}\right)_1} = \sqrt{\left(\frac{W}{L}\right)_2}\left(-\frac{\partial V_{out}}{\partial V_{in}} - \frac{\partial V_{TH2}}{\partial V_{in}}\right) \quad \text{---(11)}$$

which upon application of the chain rule

$$\frac{\partial V_{TH2}}{\partial V_{in}} = \left(\frac{\partial V_{TH2}}{\partial V_{out}}\right)\left(\frac{\partial V_{out}}{\partial V_{in}}\right) = \eta\left(\frac{\partial V_{out}}{\partial V_{in}}\right)$$

reduces to

$$\frac{\partial V_{out}}{\partial V_{in}} = -\sqrt{\frac{(W/L)_1}{(W/L)_2}}\cdot\frac{1}{1+\eta} \quad \text{---(12)}$$

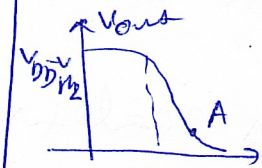
$$* \quad \sqrt{\left(\frac{W}{L}\right)_1} = \sqrt{\left(\frac{W}{L}\right)_2}\left(-\frac{\partial V_{out}}{\partial V_{in}} - \frac{\partial V_{TH2}}{\partial V_{in}}\right)$$

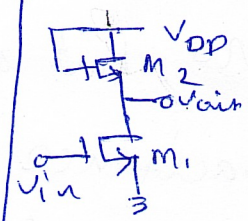
$$\text{or,} \quad \frac{\sqrt{\left(\frac{W}{L}\right)_1}}{\sqrt{\left(\frac{W}{L}\right)_2}} = -\frac{\partial V_{out}}{\partial V_{in}} - \eta\frac{\partial V_{out}}{\partial V_{in}}$$

$$= -\frac{\partial V_{out}}{\partial V_{in}}(1+\eta)$$

$$\text{or,} \quad \frac{\partial V_{out}}{\partial V_{in}} = -\sqrt{\frac{\left(\frac{W}{L}\right)_1}{\left(\frac{W}{L}\right)_2}} \cdot \frac{1}{1+\eta}$$

I/p - o/p Characteristics of a CS stage with diode connected load





It is instructive to study the overall large-signal characteristic of the circuit as well. But let us first consider the circuit shown in Fig. 4(a).

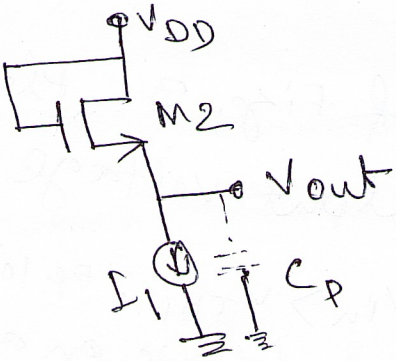
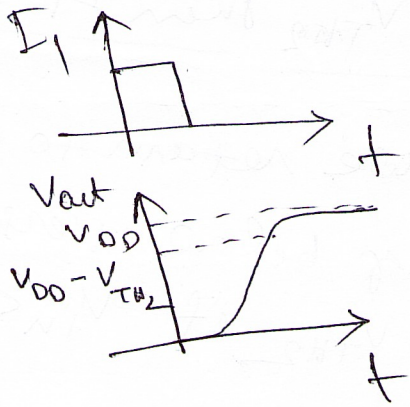


Fig 4.(a) Diode-connected device with stepped bias current



(b) variation of source voltage versus time.

What is the final value of V_{out} if I_1 drops to zero?

As I_1 decreases, so does the overdrive of M_2. Thus, for small I_1, $V_{GS2} \approx V_{TH2}$ and $V_{out} \approx V_{DD} - V_{TH2}$.

In reality, the subthreshold conduction in M_2 eventually brings V_{out} to V_{DD} if I_D approaches zero, but at very low current levels, the finite capacitance at the output node slows down the change from $V_{DD} - V_{TH2}$ to V_{DD}. This is illustrated in the time-domain waveforms of Fig 4(b). For this reason, in circuits that have frequent switching activity, we assume V_{out} remains around $V_{DD} - V_{TH2}$ when I_1 falls to small values.

Now we return to the circuit of Fig. 3. Plotted in Fig below versus V_{in}, the output voltage equals $V_{DD} - V_{TH2}$ if $V_{in} < V_{TH1}$. For $V_{in} > V_{TH1}$, Eq. 10 holds and V_{out} follows an approx. straight line.

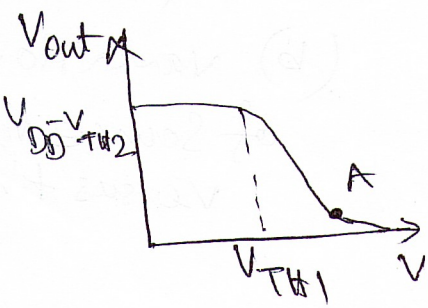


Fig. Input-output characteristic of a CS stage with diode-connected load.

As V_{in} exceeds $V_{out} + V_{TH1}$ (beyond point A), M_1

enters the triode region, and the characteristics becomes nonlinear.

The diode-connected load can be implemented with a PMOS device as well. So shown in Fig 5 the circuit is free from body effect, providing a small-signal voltage gain equal to

$$A_v = -\sqrt{\frac{\mu_n (W/L)_1}{\mu_p (W/L)_2}}, \quad \text{.....(13)}$$

where channel-length modulation is neglected

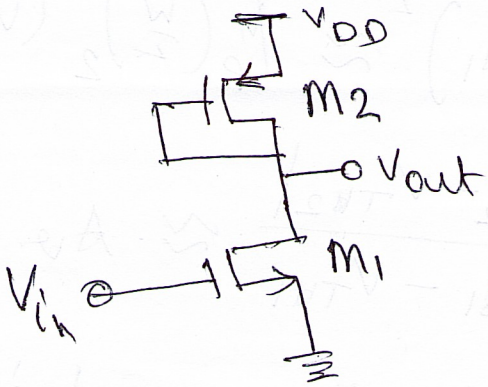


Fig. 5 CS stage with diode connected PMOS device.

Eqn (8) and (13) indicate that the gain of a common source stage with diode connected load is a relatively weak function of the device dimensions. For example, to achieve a gain of 10, $\mu_n (W/L)_1 / [\mu_p (W/L)_2] = 100$, implying that, with $\mu_n \approx 2\mu_p$, we must have $(W/L)_1 \approx 50 (W/L)_2$

In a sense, a high gain requires a "strong" input device and a "weak" load device.

In addition to disproportionately wide or long transistors (and hence a large input or load capacitance), a high gain translates to ~~one~~ another important limitation: reduction in allowable voltage swings. Specifically, since in Fig. 5,

$$I_{D1} = |I_{D2}|,$$

$$\mu_n C_{ox} \left(\frac{W}{L}\right)_1 (V_{GS1} - V_{TH1})^2 \approx \mu_p C_{ox} \left(\frac{W}{L}\right)_2 (V_{GS2} - V_{TH2})^2, \quad (14)$$

revealing that

$$\frac{|V_{GS2} - V_{TH2}|}{V_{GS1} - V_{TH1}} \approx A_v \quad \cdots \quad (15)$$

In the above example, the overdrive voltage of M_2 must be 10 times that of M_1. For example, with $V_{GS1} - V_{TH1} = 200\,mV$ and $|V_{TH2}| = 0.7\,V$, we have $|V_{GS2}| = 2.7\,V$, severely limiting the output swing. This is another example of the trade-offs suggested by the analog design octagon. Note that, with diode-connected loads, the swing is constrained by both the required overdrive voltage and the threshold voltage. That is, even with a small overdrive, the output level cannot

exceed $V_{DD} - |V_{TH}|$.

An interesting paradox arises here if we write $g_m = \mu_n C_{ox} (W/L) |V_{GS} - V_{TH}|$. The voltage gain of the circuit is then given by

$$A_V = \frac{g_{m1}}{g_{m2}} = \frac{\mu_n C_{ox} (W/L)_1 (V_{GS1} - V_{TH1})}{\mu_n C_{ox} (W/L)_2 |V_{GS2} - V_{TH2}|} \quad \cdots (16)$$

Eqn (16) implies that A_V is inversely proportional to $|V_{GS2} - V_{TH2}|$. It is left for the reader to resolve the seemingly opposite suggested by eqn (15) and (16).

We should also mention that in today's CMOS technology, channel-length modulation is quite significant and, more importantly, the behavior of transistors notably departs from the square law. Thus, the gain of the stage (CS stage with diode connected load (Fig. 3) must be expressed as

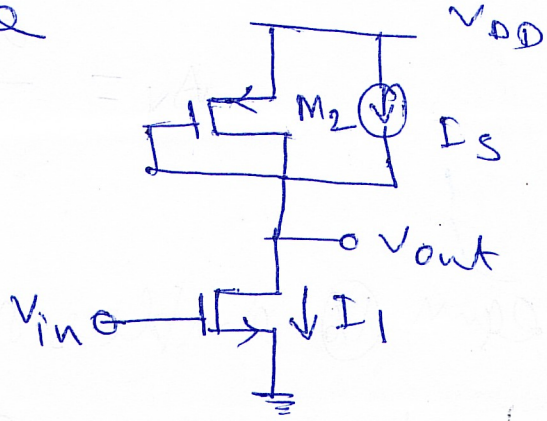
$$A_V = -g_{m1} \left(\frac{1}{g_{m2}} \| r_{O1} \| r_{O2} \right),$$

Example: In the circuit of Fig below, M_1 is biased in saturation with a drain current equal to I_1. The current source $I_S = 0.75 I_1$ is added to the circuit. How is $A_v \approx \dfrac{|V_{GS2} - V_{TH2}|}{V_{GS1} - V_{TH1}}$ modified for this case?

Soln: Since $|I_{D2}| = I_1/4$, we have

$$A_v \approx -\frac{g_{m1}}{g_{m2}}$$

$$= -\sqrt{\frac{4\mu_n (W/L)_1}{\mu_p (W/L)_2}}$$



Moreover,

$$\mu_n \left(\frac{W}{L}\right)_1 (V_{GS1} - V_{TH1})^2 \approx 4 \mu_p \left(\frac{W}{L}\right)_2 (V_{GS2} - V_{TH2})^2$$

yielding

$$\frac{|V_{GS2} - V_{TH2}|}{V_{GS1} - V_{TH1}} \approx \frac{A_v}{4}$$

Thus, for a gain of 10, the overdrive of M_2 need be only 2.5 times that of M_1. Alternatively, for a given overdrive voltage, the circuit achieves a gain four times that of the stage in Fig. 5. Intuitively, this is because for a given $|V_{GS2} - V_{TH2}|$, if the current decreases by a factor of 4, then $(W/L)_2$ must decrease proportionally, and $g_{m2} = \sqrt{2\mu_p C_{ox} (W/L)_2 I_{D2}}$ is lowered by the same factor.